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Research article

Education

# A DIDACTIC PROPOSAL FOR TEACHING SELECTED TOPICS OF COMPLEX VARIABLES IN MIDDLE EDUCATION

# 中等教育中复杂变量选题教学的教学建议

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**Abstract**

This research adopts a mixed-methods approach to explore and enhance middle-class students' understanding of complex numbers. The study delves into the relationships, applications and real-life contexts of complex numbers, considering their relevance in algebra and trigonometry. The primary objective is to design a didactic sequence enabling tenth-grade students to develop a comprehensive and geometric perspective of complex numbers, including operations, transformations, and practical applications in problem-solving and real-world situations. The research targets several key objectives. First, it identifies challenges in teaching and learning complex variables within vocational schools. Second, it demonstrates effective didactic approaches and classroom practices to enhance the complex variable instruction. Third, the study constructs a didactic proposal leveraging geometric and analytical concepts previously acquired by students in their earlier courses. Finally, the research integrates an engaging computational element to effectively validate students' learning outcomes. The study employs essential tools: the geoboard, facilitating practical knowledge acquisition, and various software applications such as GeoGebra, Excel, and the video game "Enigma" to foster student engagement and reinforce learning. This research significantly contributes to mathematics education by presenting a novel and comprehensive didactic proposal for teaching complex numbers. By addressing teaching and learning challenges, it offers valuable insights and effective classroom practices for educators. Moreover, incorporating geometric perspectives and engaging computational elements create a well-rounded and holistic learning experience for students. Ultimately, the study findings can enhance the complex variable instruction in vocational schools, positively impacting mathematics education overall.

**Keywords:** Lateral Number, Geoboard, Didactic Sequence, Didactic Engineering, Complex Number

**摘要** 本研究采用混合方法的方法来探索和提高中产阶级学生对多个理解。该研究深入研究了复数的关系，应用和现实生活中的上下文，考虑它们在代数和三角学中的相关性。主要目标是设计一个说教序列，使十年级的学生能够开发复杂数字的全面和几何视角，包括操作，转换和在解决问题和现实世界情况下的实际应用。该研究针对几个关键目标。 首先，它确定了职业学校在教学和学习复杂变量方面的挑战。其次，它展示了有效的教学方法和课堂实践，以增强复杂的可变教学。第三，该研究构建了一个教学建议，利用学生以前在早期课程中获得的几何和分析概念。最后，该研究整合了一个引人入胜的计算元素，以有效地验证学生的学习成果。该研究采用了基本工具：地球仪式，促进实用知识获取，以及各种软件应用程序，如地理代数，Excel 和视频游戏"谜，谜"，以促进学生的参与和加强学习。这项研究通过提出一个新颖而全面的教学复数的教学建议，为数学教育做出了重大贡献。通过解决教与学的挑战，它为教育工作者提供了宝贵的见解和有效的课堂实践。此外，结合几何视角和引人入胜的计算元素为学生创造了一个全面和全面的学习经验。最终，研究结果可以增强职业学校的复杂变量教学，对整体数学教育产生积极影响。

**关键词:** 横向数，地球仪式，说教序列，说教工程，复数

## I. INTRODUCTION

Taking into account the approaches of García, Mathematics in Secondary School, "the didactics of any subject means… the organization of the teaching and learning processes relevant to such subject" [1]. In this sense, didacticians are organizers, developers of education, authors of textbooks, teachers of all kinds, and even students who organize their own individual or group learning. And García continues to argue that: "didactics is the science that is interested in the production and communication of knowledge. Knowing what is being produced in a teaching situation is the objective of didactics" [1]. Thus, this proposal seeks precisely within this framework to contribute to the construction of a didactic proposal that accounts for the teaching of some topics of one of the subjects with the greatest difficulty for students and that generates great expectations regarding its mastery in our school system, since we cannot forget that it is one of the subjects on which the Ministry of National Education has recently focused, which has generated competency standards in the area of mathematics, as well as Basic Learning Rights; in its two versions, as well as fundamental contents aimed at improving student performance in this important area of knowledge. In addition, at a global level, we can highlight that researchers in this area of knowledge are currently making efforts to develop theories that develop theoretical models that contribute to improving the teaching and learning processes of mathematics, generating methodological strategies more appropriate to the reality of the student and in general to the development of research aimed at accounting for the solution to this problem. In this regard, there are concrete proposals from some specialists, such as Godino's theory of institutional and personal meanings of a mathematical object [2], Vergnaud's conceptual fields [3], Chevallard's anthropological theory of didactics mathematics [4], Brousseau's theory of didactic situations [5], Cantoral's Socio-epistemology [6], and particularly in the world of complex numbers.

In this sense, the complex variable, which allows solving equations whose solution is not within the set of real numbers, but in a dimensional extension of them, allows the calculation of all possible roots of an n-th degree polynomial; as an example, let us mention a problem posed by Diophantus (3rd century) in his work Arithmetic, where he proposes "to find the sides of a right triangle whose perimeter is 12 and area 7, a situation that leads to the equation $336\,x^2 + 24 = 172x$ , whose solution is not within the framework of real numbers, incomprehensible in a tangible way to students and their immediate environment, but whose traditional and many contemporary didactics are not resolved in learning processes. Therefore, this research proposal seeks to contribute a didactic approach that successfully accounts for the learning of some topics of a complex variables, such as complex numbers, the topology of the complex plane, operations with complex numbers, applications with complex numbers, and some basic transformations; based on the investigation of the methodologies applied in their pros and cons at the Educational Institution in Pereira, the broad reflection on them and therefore the aim at didactic innovation for the teaching-learning of one of the most incomprehensible mathematical dilemmas due to the high level of abstraction with which it is approached in young students of

basic and middle-school.

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García states that the complexity of the problems posed in the didactics of mathematics produces two extreme reactions [1]. The first are those who affirm that the didactics of mathematics cannot become a field with a scientific foundation and, therefore, the teaching of mathematics is essentially an art. In the second position, we find those who think that the existence of didactics as a science is possible and reduce the complexity of problems by selecting only a partial aspect to which they attribute special weight within the set, giving rise to different definitions and visions of it. The didactics of mathematics should tend toward what Jean Piaget [7] called transdisciplinarity, which would place research and innovations in didactics within the interactions between multiple disciplines (Psychology, Pedagogy, Sociology among others, without forgetting Mathematics itself as a scientific discipline) that allow progress in the knowledge of the problems posed [1]. For this research, the topic to be investigated is the complexity of teaching complex variable, for which the second position is assumed, since instruction in mathematics is now assumed as a scientific discipline that, based on research carried out worldwide with the aim of improving performance levels and applications thereof, responds to concerns and proposals about how it is learned, what processes are carried out in the individual's mind when working with mathematics, what are the most relevant contents for the learner, whether to resort to basic elements proposed by the Greeks or by different theories aimed at improving their understanding or whether or not it is important to use and apply technological resources, among others; which we consider to be related to various fields of knowledge such as pedagogy, epistemology, psychology, sociology, didactics, biology, etc., which ultimately support the raison d'être of mathematics as a discipline present in all scientific, technological, economic and social advances.

Although the topic of Complex Numbers is addressed in basic education textbooks in ninth grade to explain quadratic equations of the form $ax^2 + bx + c = 0$, where the discriminant $b^2 - 4ac < 0$ yields solutions outside the set of real numbers, and basic education textbooks cover the definition of complex numbers and their fundamental operations, there is a lack of focus on laying the groundwork for vocational schools' teachers to establish fundamental notions of complex variables. This is despite its importance in understanding subsequent content, its geometric applications, its relationship with trigonometry, and its significance in higher education courses and comprehending key concepts in modern mathematics.

A review of bibliographic material on teaching complex numbers at the basic education and vocational levels in the country and the region revealed the following findings: certain textbooks like "Nuevas Matemáticas 9" [8] briefly cover complex numbers, while "Matemáticas 2000" [9] emphasizes complex numbers in algebraic form and their operations. Moreover, academic works by students and faculty from the University Tecnológica de Pereira delve into the applications of complex variables in physics and complex analysis, targeting university-level students.

This study contributes novel elements to the proposal, aiming to be of great value to the educational community, specifically the Business Educational Institution and mathematics teachers in vocational schools, as well as the scientific community in the field of mathematics didactics. The proposed elements are expected to enhance the teaching and learning process of complex numbers by incorporating historical perspectives, the use of ICT tools, and fostering students'

exploration and understanding of complex numbers' geometric properties and their applications in diverse mathematical contexts.

## II. PROBLEM STATEMENT

The project will investigate the didactic problem of learning some topics of complex variables in vocational middle school, starting from the review of the didactics used by mathematics teachers, the positive and/or negative impacts on students in light of the various discourses constructed on didactics in our contemporary era applied in the international and national field, in such a way that they contribute as references for the construction of our own, but having as guiding force our own experience and contextual and cultural wisdom, and applying both qualitative and quantitative research methodology with their respective instruments for collecting and analyzing information to describe findings that allow us to successfully construct didactics in the learning of some topics of a complex variable in the vast world of mathematics. Now, the teaching-learning processes of mathematics in recent times in Colombia and the region have become one of the most complicated didactics in their teachability, since it is evidenced in many discussions, analysis of the situation, and expectations to solve this problem, particularly evidenced in the low results of national tests such as "Saber-test"s for basic education, Saber 11, and international tests such as *PISA* tests, for many students; thus developing mathematical competencies has become one of the central objectives of both the Ministry of National Education in Colombia, educational institutions, teachers and students themselves when facing various tests, university careers and even the world of work. To determine the incidence of this problem in the development of mathematical thinking among middle-class students, a diagnostic test was conducted with eleventh-grade students during the 2017 school year, which sought to determine their level of competency regarding complex numbers. The test consisted of 9 questions in which the students demonstrated low knowledge about the subject since at least 90% of the students did not answer correctly or their answers were incomplete. In the teaching field, the issue of teaching mathematics has become a great didactic debate where teachers from preschool to university blame each other for poor foundations in student training; in their training process, they must develop competencies in various thought processes aimed at learning mathematics: numerical and numbering systems, spatial and geometric systems, metric thinking and measurement systems and variational thinking and algebraic and analytical systems.

Thus, the mastery and understanding of the basic concepts of complex variables in the mathematical field and fundamentally in that of numerical sets, and in the understanding, at least intuitively, of the fundamental theorem of algebra; becomes a very useful tool since it leads to the resolution of nth degree equations, and due to their high level of abstraction they are beyond the immediate comprehension of students, but constitute the basis for the assimilation of new knowledge such as geometric knowledge related to plane movements, integrals, equations and their applications in engineering and economics. In this sense, we ask the following question; how to construct a didactic proposal for learning mathematics that contributes to minimizing the existing deficiency in competencies related to the mastery of some basic concepts of complex variables in vocational school?

In this way, the research question would be: How to contribute a didactic proposal for teaching some topics of complex variables (complex numbers, topology of the complex plane, operations with complex numbers, and some transformations) in middle school.

## III. JUSTIFICATION

Throughout history, humans have constructed various sets of numbers that agree with their needs and cultural contexts. The application of these numbers contributes to solving various problems related to their environment. In addition to mathematics, the complexity of situations requires the involvement of science, technology, and other knowledge areas related to this important tool of human thinking. In the development of their mathematical thinking at the secondary and vocational education levels, students must have a clear understanding and mastery of the properties of number sets that have been formalized. They should also be able to apply them in various contexts of their daily lives to meet the demands of solving problems in their environment. The main tool for this is the set of rational and irrational numbers, which forms the basis for the construction theory of real numbers.

In solving equations and finding the zeros of some polynomials, students often encounter solutions that lie outside the domain of real numbers. With some effort and instruction, they discover these solutions in complex numbers, which we refer to as "lateral numbers" in our proposal to facilitate understanding and avoid the

problems associated with the term "complex numbers." Although complex numbers have been used by mathematicians for centuries, their formalization was only achieved in the 20th century due to their application in various fields of modern mathematics. They are considered one of the most marvelous creations of our time, thanks to their potential in geometric, algebraic, and variational perspectives.

In an article “Polynomials in the Construction of Safe Buildings” by Dr. Jared Aurentz [10], there is a reflection on the application of complex numbers in solving equations, the application of the Fundamental Theorem of Algebra, and their use in science and the financial field through numerical analysis.

The complex variable, historically and even in modern times in the development of mathematics, is considered the core that has been able to traverse different arithmetic, geometric, algebraic, and topological methods to understand and solve many problems that remained open for centuries.

This research contributes to the strengthening of three key types of mathematical thinking: numerical, geometric, and variational. The understanding of complex variables allows for the integration of these types of thinking, supported by practical tools such as the geoboard and technological components such as Geogebra. An interactive game application will be used to enhance student learning and engagement.

This research focuses on developing a didactic proposal to make complex variables an engaging and comprehensible topic for vocational school students. It addresses the novelty and depth of this topic within mathematics education, particularly at the classroom level. The aim is to contribute elements that facilitate the teaching and learning of complex variables, making it not only a scientific concept but also an approachable topic in the field of education.



Figure 1. Grade 10B students working on the implementation of the proposal

## IV. THEORETICAL FRAMEWORK

According to Mallart, the most accurate concept of didactics is "science of learning and teaching in general" [11] and deals with normative decisions that lead to learning through the use of teaching methods. In other words, according to Mallart [11], didactics is the science of education that studies and intervenes in the teaching-learning process in order to achieve the intellectual development of the learner.

On the other hand, according to Hernández [12], didactics focuses on principles and techniques that are valid for teaching any subject or discipline, and to be valid, it must be linked to the objectives of education and centered on the learner.

According to Hernández [12], didactics are the science and art of teaching.

On the other hand, didactics is a pedagogical discipline oriented by educational purposes and committed to the improvement of all human beings through the understanding and permanent transformation of socio-communicative processes as well as the adaptation and appropriate development of the teaching-learning process.

It expands pedagogical and psycho-pedagogical knowledge by providing socio-communicative models, adaptation, and appropriate development of the teaching-learning process. However, these knowledge areas require teachers to engage in preparation and research, which elevates the task of instructional planning to a neuro-linguistic and meta-cognitive level. This leads to the definition by Medina and Salvador: “Didactics requires great reflective-comprehensive effort and the development of applied theoretical models that enable a better interpretation of the teacher's task and the expectations and interests of the students” [13].

From this perspective, didactics places learning at its core to achieve the education of learners. It also contributes to generating theories about teaching based on teaching practice in order to influence the domains of knowledge, skills, know-how, and the development of one's identity. Teachers, as didactics, are responsible for instructing, communicating, facilitating learning and imparting knowledge.

The number in mathematics is considered the most basic object and a fundamental tool in various cultures. Its use to represent quantities and magnitudes dates back to the very origin of humanity, with structured studies dating back over five thousand years The different numerical sets used today have gone through stages ranging from the use of various symbols to the acceptance of additive or positional numbering systems developed by ancient cultures such as the Babylonians, Egyptians, Greeks, Romans, Chinese, Arabs, Hindus, and indigenous cultures. Even today, cultures such as the Mayans and

Embera-Chamí in Colombia continue to use their own numeral systems. Additionally, numerical sets such as complex numbers have been formalized and widely used in the scientific community. The origin of most numbering systems is related to the need for counting, classifying, and numbering, primarily using parts of the body such as fingers. Each number or mathematical object in these numbering systems is considered a numeral and constitutes the fundamental building block of mathematics, with arithmetic, according to Gauss, being the queen of the sciences.

According to Hernández, in mathematics, the definition of number extends to include abstractions such as fractional, negative, irrational, transcendental, and complex numbers [14]. However, in some scientific circles, more abstract types of numbers, such as hypercomplex numbers, hyperreal numbers, or quaternions, are used and accepted. In Western culture and practically universally, the most well-known and used numbers are the Real Numbers, which generalize and provide solutions to many problems. However, real numbers cannot solve certain equations or find roots of all polynomials, which is why they are not considered algebraically closed. This necessitates the introduction of complex numbers, which can be seen as an extension of the set of real numbers. Complex numbers are a subset of the complex set and constitute the smallest algebraically closed field that allows finding zeros of polynomials with real variables, providing solutions to many mathematical problems and problems in other sciences. In the set of real numbers, some numbers are not solutions to a polynomial equation with rational coefficients and can be expressed as a Cauchy sequence. Examples of such numbers are $\pi$ and $e$, which are widely used in measuring curve magnitudes and in the calculation of natural logarithms. They are known as transcendental numbers and are not roots of any algebraic equation with non-zero integer coefficients.

## A. Complex Numbers

A complex number is a number of the form $z = x + iy$, where $x$ and $y$ are its real and imaginary parts. The set of these numbers is denoted by $\mathbb{C}$: $\mathbb{C} = \{ + iy : x, y \in \mathbb{R}, i^2 = -1\}$. Similarly, they can be defined as an ordered pair $(x, y)$ of real numbers, where the real and imaginary parts are given by: $x = \Re e\ z$; $y = \Im m\ z$; $z = \Re e\ z + \Im m\ z$.

The equality, addition, and multiplication as well defined and the set of complex numbers $\mathbb{C}$, together with the operations of addition and scalar multiplication by a real or complex number, form a field.

Complex numbers can be represented in various forms: rectangular (Cartesian), polar, exponential (Euler's formula), vector, and matrix. In rectangular form, a complex number is expressed as the sum of a real part and an imaginary part $(a + bi)$. The polar form represents a complex number using its magnitude and argument $\left(r(cos\theta + isin\theta)\right)$ the exponential form relates the exponential, cosine, and sine functions $\left(re^{i\theta}\right)$. Vector form treats a complex number as a two-dimensional vector in the complex plane. The matrix form represents a complex number as a matrix $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$.

The concept of transformation can be defined as a correspondence between elements of sets. It is a rule or association that assigns a unique element of set B to each element of set A, which can be represented: $A \to B, x \to y = f(x)$. If the variable $x$ represents any complex number, a function (transformation) with a complex variable is called a complex-variable function.

A transformation or complex-variable function, according to González [15], is defined as a rule that assigns a complex number $w \in \mathbb{C}$ to each $z \in D$ in a set $D \subset \mathbb{C}$.

$$f: D \subset C \to C,\ z \to f(z) = w \qquad (1)$$

Here, $D$ represents the domain of transformation. $w = f(z) = u(x,y) + iv(x,y)$, where $z = x + i\,y$, $u = \Re e f(z) = \Re e\, w$, and $v = \Im m\, f(z) = \Im m\, w$ (Fig. 2).

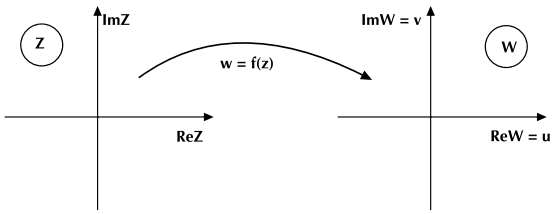



Figure 2. Complex argument function

## B. Inverse Transformation

The inverse transformation or inversion mapping is given by $w = \frac{1}{z}$, $z \neq 0$. This transformation has the following effects:

- Circles in the plane $P$ are transformed into circles in the plane $P'$.
- Circles passing through the origin in $P$ are transformed into lines in the plane $P$'.
- Lines not passing through the origin in the plane $P$ are transformed into circles passing through the origin in the plane $P'$.
- Lines passing through the origin in the plane $P$ are transformed into lines passing through the

origin in $P'$.

### C. Quadratic Transformation

It is a transformation of the form $w = f(z) = \alpha z^2 + \beta z + \gamma$, where $\alpha, \beta, \gamma \in \mathbb{C}$ and $\alpha \neq 0$.

The most elementary case of this transformation is given by $w = f(z) = z^2 = x^2 - y^2 + 2xy\, i$.

For example, the complex number $z = 2 + i$ is transformed into the number $w = 3 + 4i$.

Furthermore, this transformation maps lines in plane P to curves (parabolas) in plane P' (Fig. 3).

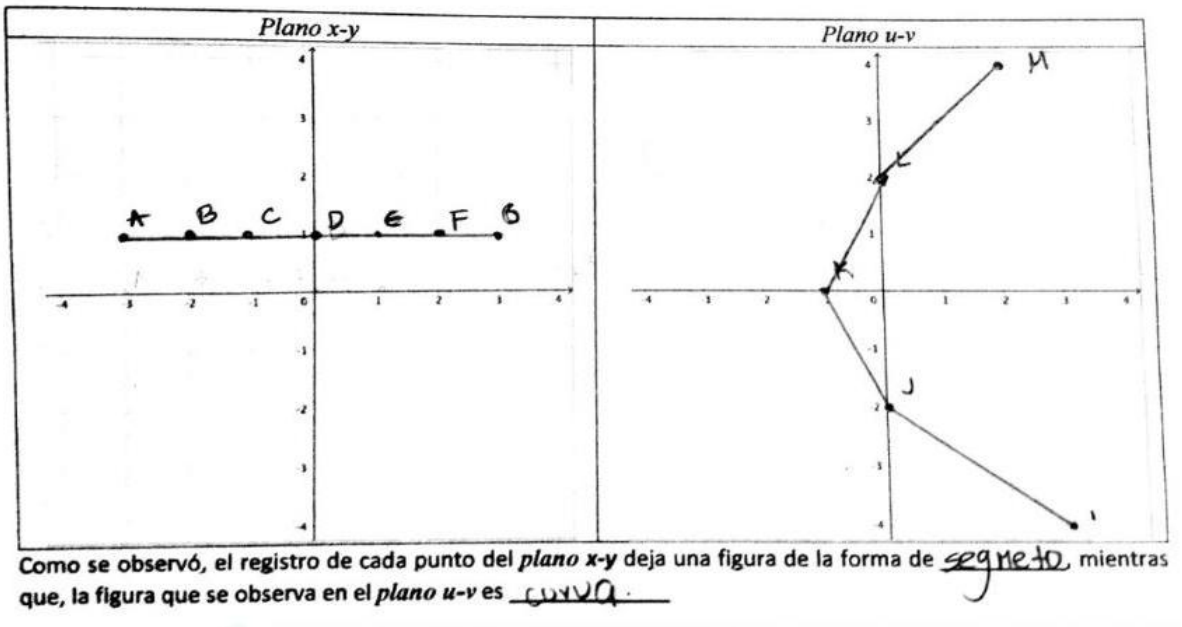



Figure 3. The photo of a student's response about quadratic transformation

## V. Didactic Engineering

The notion of didactic engineering was introduced in French mathematics education in the early 1980s. According to Artigue, "This term refers to a form of didactic work comparable to the work of an engineer who, to perform a specific project, relies on the scientific knowledge of their field and accepts scientific-type scrutiny" [16].

### A. General Characteristics

According to Artigue, "Didactic engineering is characterized, first and foremost, by an experimental framework based on didactic sequences carried out in the classroom, i.e., on the design, implementation, observation, and analysis of these sequences. Generally, two levels are distinguished: micro-engineering and macro-engineering" [16].

Furthermore, De Faria mentions: "The term didactic engineering is used in mathematics education with a dual function: as a research methodology and as the production of teaching and learning situations" [17].

This idea is supported by researchers Calderón and León, who consider that: "A fundamental characteristic of this research methodology is the confrontation between a priori analyses of classroom activity designs and A posteriori analyses that are produced in the implementation of tasks, as the basic form of validation for the hypotheses described in the research" [18].

### B. Phases of the Didactic Engineering Methodology

This research methodology consists of four phases.

*1) Preliminary Analysis Phase*

In this phase, the focus is on deepening the understanding of the epistemological analysis of the content covered in teaching, analyzing traditional teaching and its effects, examining the conceptions, difficulties, and obstacles that influence evaluation, and determining the constraints in which the didactic action will occur. According to Artigue et al.: "Preliminary studies only maintain their preliminary quality at their initial level of elaboration" [16]. Subsequently, these studies take on different positions and functions in the research according to the specific objectives of the investigation.

*2) Conception and A Priori Analysis of Didactic Situations Phase*

Traditionally, this a priori analysis consists of a descriptive and predictive part, focusing on the characteristics of a non-didactic situation that has been designed and will be proposed to students. It describes the local choices (relating them to global ones) and the characteristics of the

didactic situation that emerge from them. It analyzes what a student could learn in this situation based on the possibilities of action, decision, control, and validation available to them when working independently of the teacher. Possible behaviors are anticipated, and it is sought to demonstrate how the analysis enables controlling their meaning and ensuring that if the expected behaviors occur, they result from the implementation of the intended learning.

Regarding this, Godino et al. conceptualize: "This a priori analysis should be conceived as an analysis of sense control. In a simplified manner, if the constructivist theory establishes the principle of student participation in the construction of their knowledge through interaction with a specific environment, the Theory of Didactic Situations, which serves as a reference for the didactic engineering methodology, has aimed, from its inception, to become a theory of the control of relationships between sense and situations" [19].

*3) Experimentation Phase*

This phase of didactic engineering involves direct contact between the researcher and a specific population of students who are the subjects of the research. In this phase, the researcher must clearly explain the objectives and conditions of the research to the participating students, establish a didactic contract, and finally, apply the research instruments designed based on the research problem.

*4) A Posteriori Analysis and Evaluation Phase*

This phase is based on the collection of data throughout the experimentation process, including observations of teaching sequences and student productions in and outside the classroom. These data are supplemented with additional information obtained through external methodologies such as questionnaires and individual or small group interviews conducted during each teaching session. The validation or refutation of the hypotheses described in the research is based on the comparison of the a priori and A posteriori analyses, allowing for a comprehensive evaluation [18].

## C. The Geoboard

In this research, the geoboard in its various forms such as square, circular, triangular, unit circle, or trigonometric circle is considered a valuable didactic tool. It not only allows students to manipulate geometric shapes but also provides opportunities for consolidating existing knowledge, acquiring new knowledge, and especially contributing to the development of abstract thinking through activities based on concrete operations and inductive reasoning, as proposed by Piaget in his cognitive theory.

## D. Overview of the Geoboard

The geoboard is a didactic tool used to introduce and reinforce concepts of plane geometry [20]. It consists of a board with a grid of squares and nails or pins inserted at each corner (Fig. 4). Colored rubber bands are stretched and secured on nails to create geometric figures. The geoboard facilitates the teaching and learning of geometry, including angles, polygons, similarity, symmetry, and more. It integrates content from various mathematical areas such as geometry, arithmetic, algebra, trigonometry, and calculus. It can be used to represent complex numbers visually on the plane and explore their properties and operations. The geoboard also allows exploration of geometric transformations and the relationship between complex numbers and the unit circle. Overall, the geoboard provides pedagogical opportunities for hands-on activities and visual representations to deepen students' understanding of complex numbers and their applications in mathematics and other disciplines.

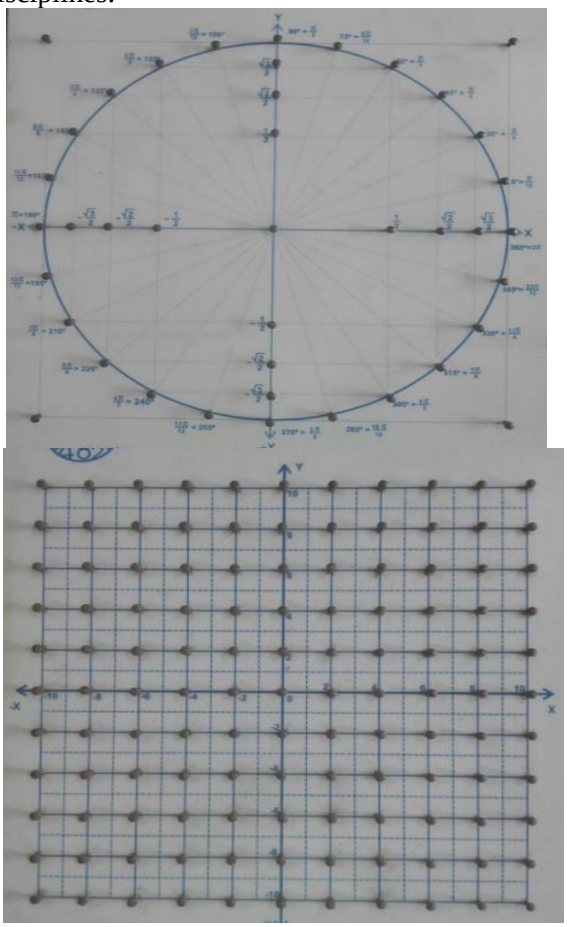


Figure 4. Photo of the circle and the square geoboard

In addition, the use of the square geoboard promotes collaborative work among the tenth-

grade students of the Educational Institution. Not only did they use it as a didactic tool, but they also constructed the geoboards themselves with the help of their classmates and parents.

The use and implementation of information and communication technologies (ICTs) in education have become a pervasive reality across various levels, including economic, social, political, cultural, and educational spheres. Despite the continuous evolution of hardware and software, we often lag behind in fully harnessing the immense possibilities offered by ICTs, both in recreational and educational contexts. In the field of mathematics, the Ministry of Education (MEN) introduced Curriculum Guidelines in 1999 [21] to promote the continuous use of ICTs as mediators between the mathematics curriculum and students in the classroom. However, the underutilization of ICTs persists due to teachers' fear of technology replacing their roles, limited access to technological resources, and students not fully embracing ICTs as tools for cognitive enhancement and the development of crucial mathematical thinking processes such as reasoning, problem-solving, communication, modeling, and procedure-based activities. Nonetheless, recognizing the potential of ICTs is vital in empowering students and fostering their engagement in mathematics education.



Figure 5. The photo of the students working with the geoboards

The use of technology in the teaching and learning process of mathematics has multiple benefits. Not only does it facilitate the acquisition of knowledge in a creative, dynamic and innovative way but it can also be motivating for students. Devices such as tablets and smartphones are widely used by students in their everyday lives and should be leveraged as learning tools. According to cited by Suazo, the use of educational software is fundamental in this regard [22]. Integrating technology in the teaching of mathematics not only enriches the learning process but also prepares students for the digital world we live in, providing them with relevant skills for their future development.

These functions allow us to clearly understand the roles that both students and teachers play in using ICTs in the implementation of the mathematics curriculum and the evaluation of its contents. Hence the importance of the Enigma game, which relies on the acquired knowledge of lateral numbers for participation and enjoyment. The use of Excel, a widely used spreadsheet software that students are familiar with, as well as Geogebra, a free educational software with great applicability for teaching and learning Geometry in a dynamic and active way, enabled the use of technology for implementing the proposal.

### E. Likert Scale

The Likert scale is named after the American organizational psychologist Likert (1903-1981). It is a psychometric scale used in research. The scale is employed to analyze questionnaires, where the respondent has various alternatives to indicate their degree of maximum satisfaction or dissatisfaction, including an intermediate option. Typically, the number of response alternatives is an odd number. At the end of the scale, it allows for both vertical and horizontal analysis of the responses. According to Malavé [23], the Likert scale can have the following alternatives: 5 - Strongly Agree; 4 - Agree; 3 - Neither Agree nor Disagree; 2 - Disagree; 1 - Strongly Disagree. In this research work, a Likert scale is used to perform both horizontal and vertical (quantitative and qualitative) analyzes of the evaluation results conducted with the tenth-grade students to assess their mastery and understanding of complex numbers, also known as lateral numbers.

## VI. Methodological Design

### A. Mixed Research Approach

Considering that the objective of this research is to develop a didactic proposal for teaching lateral numbers to tenth-grade students at the Educational Institution, it corresponds to a mixed-method research, as proposed by Echevarría [24]. The presentation we have developed highlights the significance of using mixed research methods, including the different classifications within social sciences and their application in research analysis, as provided by Echevarría [24] and Pereira [25].

These authors, among others, demonstrate the great potential of going beyond the scope and limitations of quantitative and qualitative research methods by opting for a mixed-method approach. This approach allows for partial integration or complete integration of methods, offering multiple possibilities to account for the

results of research conducted under this modality.

These possibilities are expressed through triangulation, which involves comparing and using both quantitative and qualitative methods to generate results that are not commensurable. This approach is questioned, and complementation comes into play, aiming to develop, enhance, illustrate, and clarify the results of one method with the results of another method, depending on how one method is subsumed under the other. Combination, on the other hand, involves integrating these methods to strengthen the validity of the research by compensating for the weaknesses of each method. This represents the strong mutual support that both methods provide when presenting research results (Tables 1 and 2).

Triangulation of different methodologies is a research strategy used to obtain a more comprehensive and robust understanding of a phenomenon. It involves using multiple methodological approaches, such as quantitative, qualitative, and mixed methods and combining different data collection and analysis techniques [24].

Table 1.
Triangulation of different methodologies

| **Logical Approaches** | **[26]** | **[27]** |
|---|---|---|
| The independence of methods; convergence, results confrontation | Triangulation Initiation | Triangulation |
| Method collaboration; method dependency, modification of one by the other | Development | Combination |
| Method collaboration result complementation, independence of methods (one modifies the others) | Complementary expansion | Complementation |

Table 2.
The matrix of mixed methods and components (Adapted from [24])

| **Type** | **Monocomponent** | **Multicomponent** |
|---|---|---|
| Monomethod - MM | Cell 1:<br>Monocomponent monomethod designs<br>Traditional quantitative design<br>Traditional qualitative design | Cell 2:<br>Multicomponent monomethod designs<br>Parallel monomethod<br>a. QUAN+QUAN<br>b. QUAL+QUAL<br>Sequential monomethod<br>QUAN->QUAN<br>QUAL->QUAL |
| MM | Cell 3:<br>Monocomponent quasi-mixed design<br>Monocomponent conversion design | Cell 4:<br>Multicomponent MM designs<br>Parallel mixed design<br>Sequential mixed design<br>Conversion mixed design<br>Multilevel mixed design<br>Fully integrated mixed design<br>Multicomponent quasi-mixed design |

Consequently, many educational problems are often addressed through this mixed methodological approach to research, such as learning processes, teaching, and educability didactic strategies, characteristics of teacher educators, students, educational climate, types of curriculum content, opinions about teachers and types of teachers, academic progress of students, school performance, efficiency, and school environments, among many aspects of the educational world.

This research is guided by the principles of the mixed method described above. First, we approach the teaching and learning phenomena of basic concepts of complex variables from the perspective of didactic categories such as numbers, complex numbers, geoboards, and didactic engineering. The focal group for analysis is the 10B grade students at the Business Educational Institution in the municipality of Dosquebradas, Risaralda.

Second, we employ quantitative data collection techniques such as surveys (for teachers and students), pre-tests, and post-tests to obtain statistical information about the methodology used in teaching the basic elements of complex variables. These quantitative data are then triangulated with the aforementioned analytical categories through instruments such as the Likert scale, allowing us to identify various issues such as methodological orientation, and students' difficulties in approaching problems that involve the application of complex number concepts and transformations, among other aspects that will be further elucidated. This process facilitates the development of an

alternative didactic proposal for teaching and learning complex numbers, based on the geometric aspect and the use of technologies and concrete materials such as various geoboards.

The design of this research is based on Didactic Engineering, which is grounded in the theory of didactic situations proposed by Guy Brousseau [5]. This approach is relevant when investigating how students acquire knowledge and allows the researcher to develop their own didactic teaching sequences flexibly to mediate the understanding of mathematical knowledge that they aim to enhance in students.

**B. Summary of the Four Phases**

The didactic engineering and activities to be carried out include several phases.

*1) Phase 1: Preliminary Analysis*

*Objective:* To understand the historical, epistemological, didactic and cognitive aspects involved in teaching lateral numbers.

*Activities:*

- Literature review of the main epistemological and didactic difficulties teaching lateral numbers;
- Analysis of teachers' conceptions regarding the teaching of lateral numbers;
- Analysis of students' conceptions regarding the learning of lateral numbers.

*2) Phase 2: Conception and A Priori Analysis*

*Objective:* Identify the difficulties that students encounter when solving problems and situations in their environment that involve the application of concepts related to lateral numbers.

*Activities:*

- Design, construction, and development of a didactic situation focusing on applied problems related to complex variables;
- Analysis of the results of the didactic situation to identify the main difficulties students face in applying the concepts related to complex variables.

*3) Phase 3: Experimentation*

*Objective:* Develop and analyze a didactic proposal for the mastery and understanding of the basic knowledge of complex variables.

*Activities:*

- Design, development and implementation of didactic situations associated with complex variables;
- Analysis of students' comprehension achieved through the development of didactic situations.

*4) Phase 4: A Posteriori Analysis and Validation*

*Objective:* Validate the level of student learning about complex variables through their participation and enjoyment of the game Enigma.

*Activities:*

- Analysis of data obtained throughout the experimentation.
- Comparison of the a priori and a posteriori analyzes of the levels of comprehension achieved by students regarding complex variables.

*Preliminary analysis:* The methodology of didactic engineering consists of four phases, and according to Artigue [16], it considers three dimensions in its development: the epistemological dimension, the cognitive dimension, and the didactic dimension.

*Epistemological analysis of complex variable:* In this section, we will explore the historical development, obstacles faced, significant moments in societies that discussed it, and the motivations that drove the study of complex variables. The aim is for students and teachers to appreciate its historical evolution and incorporate it into their academic and intellectual formation.

**C. The Short History of Complex Numbers**

Complex numbers are considered a remarkable creation of the 19th-century and have become a fruitful branch of mathematics. The history of complex numbers is closely tied to the development of algebra, the solution of cubic and quartic equations, and linear algebra. The existence of complex numbers was mentioned in ancient times, with mathematicians like Diophantus in the 3rd-century using them to solve certain problems. The concept of imaginary solutions emerged through the works of mathematicians like Tartaglia and Cardano in the 16th century, who recognized the significance of numbers with negative square roots. René Descartes introduced the term "imaginary numbers" and proposed geometric methods for solving equations involving complex numbers. Euler further advanced the understanding of complex numbers and introduced the symbol "i" for the square root of -1. The concept of representing complex numbers as points in the complex plane was established by Argand, and Gauss formalized complex numbers as a numerical set, defining their properties and operations. Mathematicians like Riemann and Cauchy contributed to the development of complex variables and their applications in various fields including engineering and study of fractals.

Complex numbers characterized by the form $z = a + bi$, where $a$ and $b$ are real numbers, have a rich history intertwined with the evolution of mathematics. Their origins can be traced back to ancient times, with early mathematicians recognizing the need for numbers with negative

square roots to solve certain problems. The term "imaginary numbers" was introduced by Descartes, who also proposed geometric methods for their manipulation. Euler made significant contributions by introducing the symbol "i" and performing operations with complex numbers. Argand and Gauss further refined the understanding of complex numbers, with Gauss formalizing them as pairs of real numbers and introducing key concepts such as conjugates, addition, subtraction, multiplication, division, powers, and roots. The concept of complex variables and their applications in diverse fields was developed by mathematicians such as Riemann, Cauchy, and others. Complex numbers also find relevance in the study of fractals, which challenge traditional notions of integer dimensions. The evolution of complex numbers reflects the continuous growth and exploration of mathematics over centuries (Fig. 6-7).

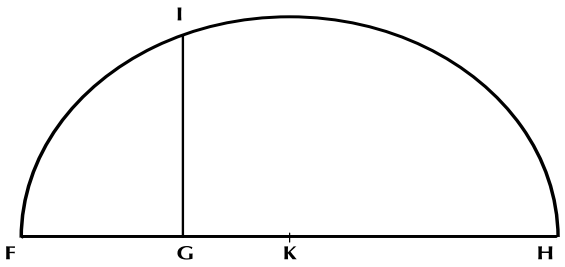



Figure 6. René Descartes introduced a geometric interpretation for extracting square roots of negative numbers

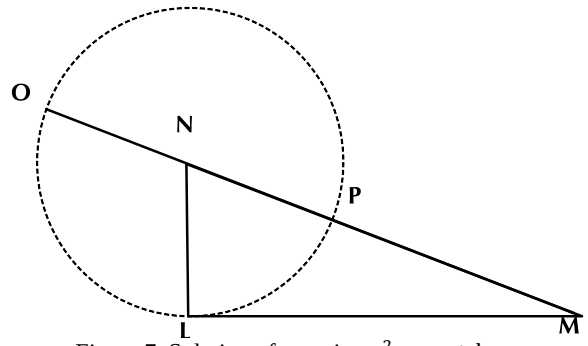



Figure 7. Solution of equation $z^2 = az + b$

A fractal is a geometric artistic design whose basic structure repeats at different scales. In many cases, fractals can be generated by a recursive process capable of producing self-similar structures independent of the specific scale, as in the case of Mandelbrot fractal. According to Du Satoy [28], the scaling properties of fractal shapes define their dimensions, which can include decimal quantities. For example, the mathematical coastline of Great Britain has a fractal dimension of 1.26, and other forms in space such as broccoli or cauliflower exhibit fractal dimensions ranging from 2 to 3. These ideas challenge the traditional paradigm of integer dimensions (a line: one dimension, a plane: two dimensions, a cube: three dimensions) [28].

For the construction of beautiful fractals today, ICT (Information and Communication Technologies) can be used, specifically software such as Geogebra, along with a basic understanding of the complex plane, its square powers, and transformations. Fractals are obtained by iterating square powers of points inside and on the boundary of the unit circle, which are the points that converge, and adding a constant C, which is another complex number. This process creates captivating figures. Moreover, these figures become interactive actions that students can explore, showcasing one of the most evident applications of the potentiality offered by understanding this mathematical object, as illustrated by the following Fig. 8.

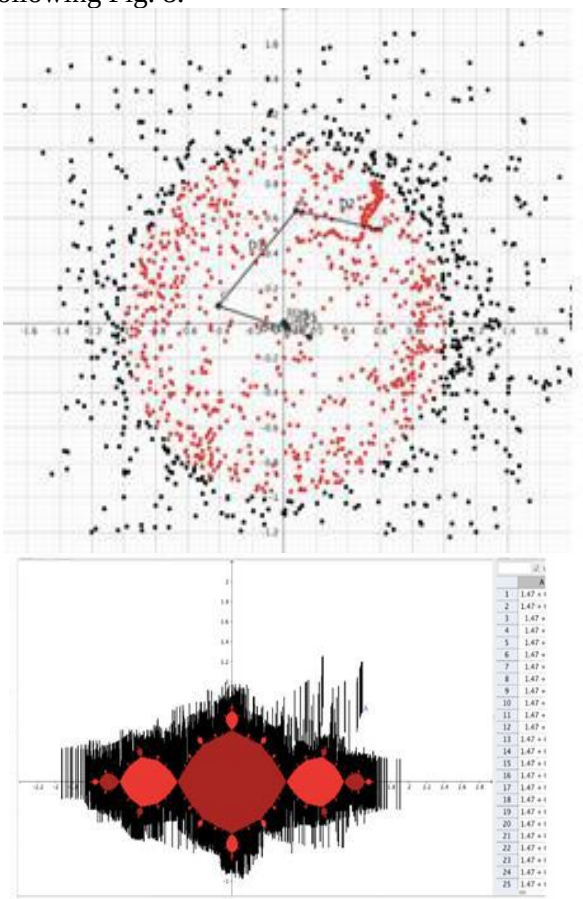


Figure 8. Fractal construction by students group in geogebra

**D. Morning Session - Teacher One**

This teacher has an experienced teacher with over twenty-five years of teaching experience and successfully completed the exercise. However, he mentioned that the topic is not covered in the school curriculum due to time constraints and the fact that it is excluded from the curriculum.

**E. Afternoon Session - Teacher Two**

This teacher pursuing a master degree in Mathematics Education with more than ten years of teaching experience also successfully completed the exercise. He emphasized the importance of addressing the topic despite its absence from the curriculum, as it offers great potential for enhancing students' levels of education.

A survey was also conducted among twenty mathematics teachers from various schools in Colombia. The results indicate that the topic of fractals is not extensively covered in most educational institutions, there is no specific methodology for teaching it, and therefore, many teachers are unaware of its significance in strengthening mathematical thinking.

**F. The Analysis of Pre-Test Results**

In this part of the research, two exercises were developed as non-didactic situations. These exercises were designed in groups of three students within the classroom and are described as follows.

*1) City Mapping*

The aim of this exercise was to observe how students approached a problem involving the location of various places in Pereira's city on a Cartesian plane.

*2) Point Operations and Geoplane Transformations*

The second exercise involved performing operations such as addition, subtraction, multiplication and division with points on a rectangular geoplane. It also included transformations of a triangle.

These exercises were intended to assess students' understanding and skills in relation to the specific topics covered. The results of the pre-test will provide valuable insights into students' prior knowledge and help inform the design of instructional strategies for teaching fractals in a more comprehensive and effective manner.

Fig. 9 illustrates the response provided by the students to the non-didactic situation, where they were presented with a contextual problem regarding the location of certain places in the city of Pereira.



Figure 9. The participation of students in non-didactic situations

The students have a clear understanding of the location of specific places in the city, which correspond to points on the plane.

The students accurately identified the coordinates of the given places, demonstrating their proficiency in understanding and applying the concepts of Cartesian coordinates. They successfully connected the real-world context with the mathematical representation on the plane.

Their ability to map the locations of the city indicates their spatial reasoning skills and understanding of coordinate geometry. The students showed competence in interpreting and using the coordinate system to represent real-life situations.

It was confirmed that the students actively engaged with the problem and applied their knowledge effectively. Their responses demonstrate their grasp of the concept and their capability to transfer mathematical ideas into practical scenarios.

This level of clarity and accuracy in locating the places within the city reflects the student's understanding of the relationship between mathematics and the real world. This indicates that they have developed spatial awareness and the ability to apply mathematical concepts to everyday situations. This demonstrates that the student lacks clarity regarding the appropriate tool to use in finding the value, namely the Pythagorean theorem. Therefore, the use of didactic strategies that reinforce the topic, as proposed in this research, can lead the student to experience cross-curricular and contextual learning.

The results of the first exercise are summarized in table 1, which presents a Likert-type analysis of the students' responses. A weighting scale ranging from “excellent” (assigned a value of 5) to “very poor” (assigned a value of 1) is used to evaluate the students' performance. The analysis assesses the correctness of the procedures employed by the students and whether they arrived at the correct answer.

By analyzing the data, we can observe the distribution of students' performance across the Likert scale. It provides insight into their understanding and proficiency in solving the given problems. This analysis allows us to identify areas where students excelled and areas where further instruction and support may be needed.

It is important to note that the lower ratings on the Likert scale indicate areas where students struggled or did not attempt to solve the questions. This highlights the need for targeted

instructional strategies to address identified misconceptions or gaps in understanding.

In conclusion, the results of the Likert analysis shed light on the students' proficiency in the given exercise. It highlights the areas of strength and areas requiring further attention, emphasizing the importance of using effective didactic strategies to enhance students' understanding and application of mathematical concepts.

Fig. 10 presents the results of the survey conducted with the students. The survey consisted of eleven questions (P1-P11), and each question was rated on a Likert scale from 1 to 5. The scores for each group (G1-G11) are shown along with the average score for each group.

Based on the results, it can be observed that Group G1 and Group G2 had the highest average scores of 3.18, indicating a relatively higher level of proficiency in the surveyed topics. On the other hand, Group G6 had the lowest average score of 1.82, suggesting a lower level of understanding.

| Group | Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 | Q9 | Q10 | Q11 | Average score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| G1 | 5 | 2 | 3 | 5 | 5 | 5 | 1 | 1 | 4 | 3 | 1 | 3,18 |
| G2 | 5 | 5 | 5 | 5 | 5 | 2 | 1 | 1 | 3 | 2 | 1 | 3,18 |
| G3 | 5 | 2 | 5 | 2 | 2 | 5 | 1 | 1 | 3 | 3 | 1 | 2,73 |
| G4 | 1 | 3 | 3 | 4 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 2,09 |
| G5 | 3 | 3 | 3 | 3 | 1 | 5 | 2 | 2 | 2 | 1 | 1 | 2,36 |
| G6 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 1,82 |
| G7 | 3 | 1 | 3 | 4 | 2 | 5 | 2 | 2 | 2 | 1 | 1 | 2,36 |
| G8 | 3 | 2 | 3 | 3 | 2 | 4 | 2 | 2 | 2 | 1 | 1 | 2,27 |
| G9 | 3 | 3 | 3 | 3 | 2 | 3 | 2 | 2 | 2 | 1 | 1 | 2,27 |
| G10 | 3 | 2 | 3 | 2 | 3 | 5 | 2 | 2 | 2 | 1 | 1 | 2,36 |
| G11 | 5 | 5 | 5 | 3 | 2 | 3 | 2 | 2 | 2 | 1 | 1 | 2,82 |
| Group average | 3,45 | 2,7 | 3,4 | 3,2 | 2,5 | 3,7 | 1,7 | 1,7 | 2,3 | 1,45 | 1,0 | 2,50 |

Figure 10. Survey results for the student questionnaire (Q)

Looking at the individual question scores, it is evident that some questions received higher ratings across all groups, indicating a better grasp of those particular concepts. For example, questions P1, P4, P5, and P6 received higher scores, suggesting that students generally performed well in those areas.

However, questions such as P2, P3, P7, and P8 received lower scores, indicating areas where students struggled or had misconceptions. These questions may require further instruction and support to improve students' understanding and performance.

The average group scores provide an overall picture of the students' performances. Groups with higher average scores tend to have a better understanding of the surveyed topics, while groups with lower average scores may require additional attention and targeted interventions.

It is important to consider these survey results in conjunction with other forms of assessment and classroom observations to gain a comprehensive understanding of students' mathematical abilities and tailor instructional strategies accordingly.

Overall, the survey results provide valuable insights into students' performance and can inform instructional planning and interventions to enhance their understanding and mastery of mathematical concepts (Fig. 11).

Regarding the average scores obtained for each of the questions posed to the students, it was observed that only four out of the eleven questions received a score of three. The highest score achieved was 3.5 (Question 6), while the lowest score was 1.0, corresponding to Question eleven (Table 4).

Table 4.
Comparative analysis of exploration activity and final test

| Knowledge | Initial Test | | Final Test | |
|---|---|---|---|---|
| | Approved Questions | Questions in Progress | Approved Questions | Non-Approved |
| Recognize lateral numbers as an ordered pair of points in the plane and use Circular and Rectangular Geoboards | 27% | 73% | 90% | 10% |
| Recognize radius and angle as elements of a lateral number in angular or polar form | 0% | 100% | 48% | 52% |
| Solve addition, subtraction and multiplication operations with points in the plane | 0 | 100% | 76% | 24% |
| Apply lateral numbers in angular form to solve real-life problems | 18% | 72% | 48% | 52% |
| Determine the effects of transformations when operating on points in a region, such as translations, dilations, and rotations | 25% | 75% | 72% | 28% |
| Apply transformations to solve everyday problems | 25% | 75% | 54% | 46% |
| Perform geometric partitions of unity | 0% | 100% | 68% | 32% |

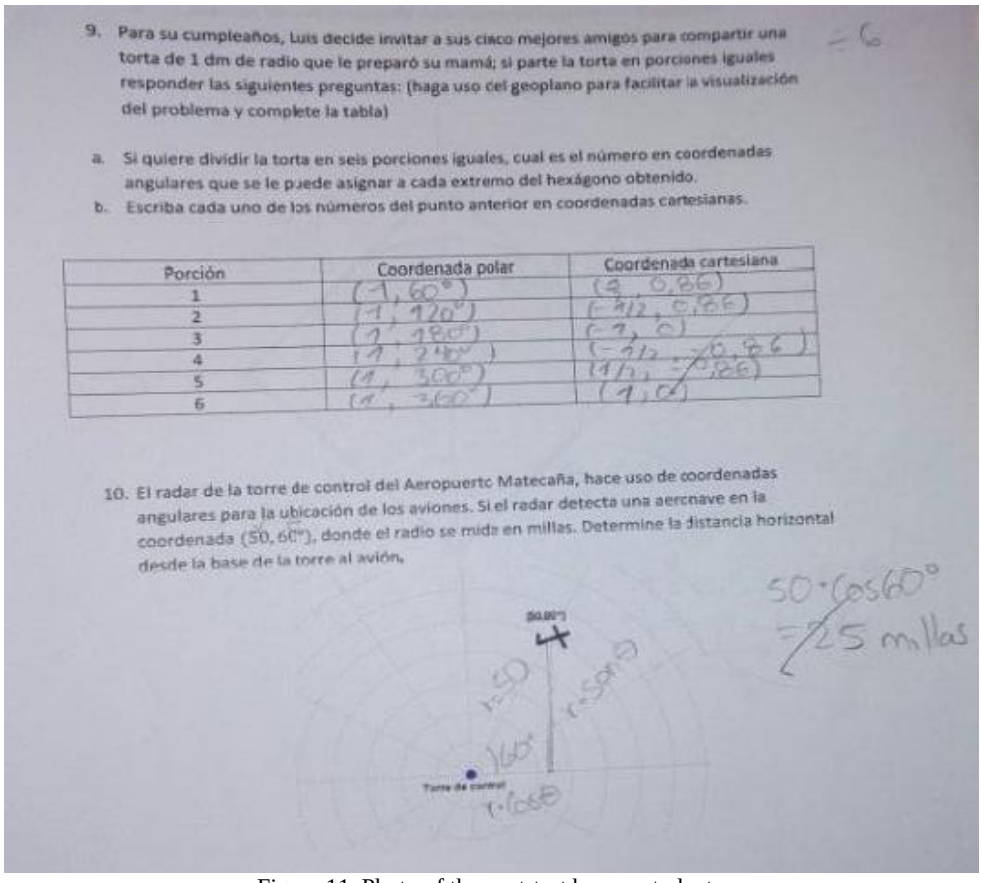


Figure 11. Photo of the post-test by one student

From the analysis based on the Table 4, we can mention the following general findings regarding the students in our research:

1. They demonstrate a better ability to recognize and establish the importance of lateral numbers as ordered pairs to perform operations and apply them in solving everyday situations.

2. They can identify lateral numbers in alternative representations, such as angular or polar and exponential forms.

3. They can perform addition, subtraction, multiplication and division operations with lateral numbers.

4. They can apply lateral numbers to solve problems within their environment, including simple optimization problems, radar problems, and GPS-related problems.

These findings suggest that the students have developed a solid understanding of lateral numbers and their applications, particularly in real-life scenarios.

The students were challenged to perform transformations of figures in the plane using the concepts of translation, rotation, and reflection (Fig. 12, 13).

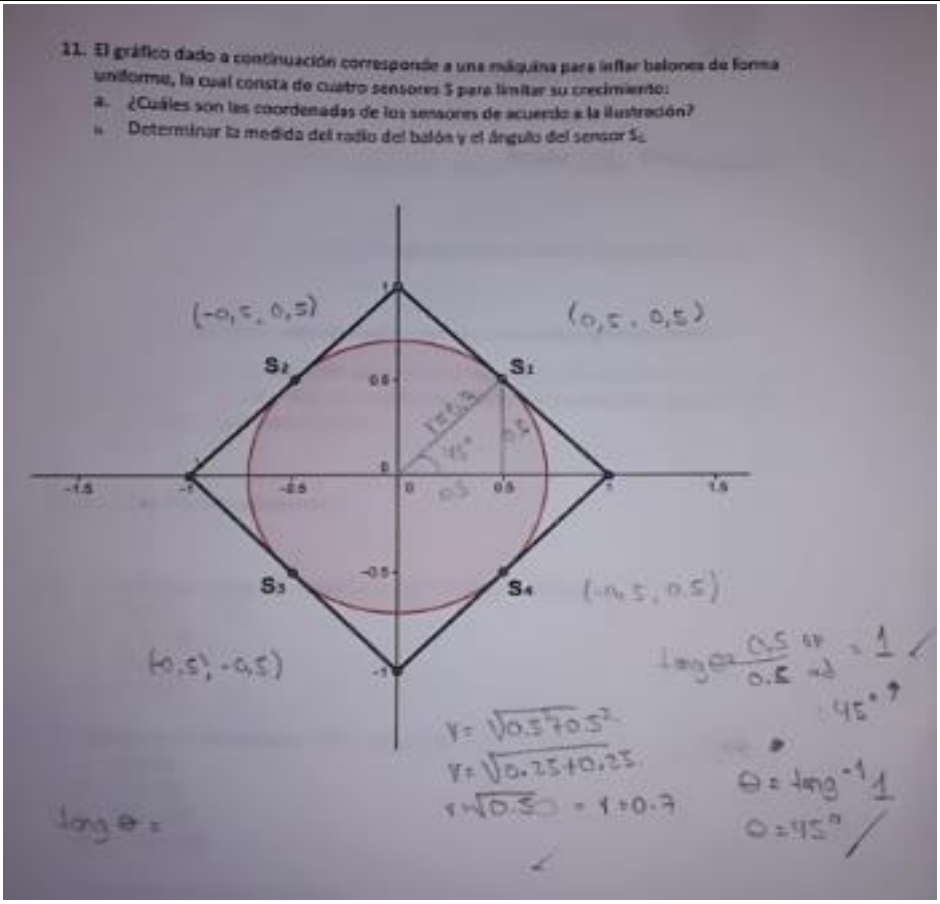


Figure 12. The photo of the answer to Question 11

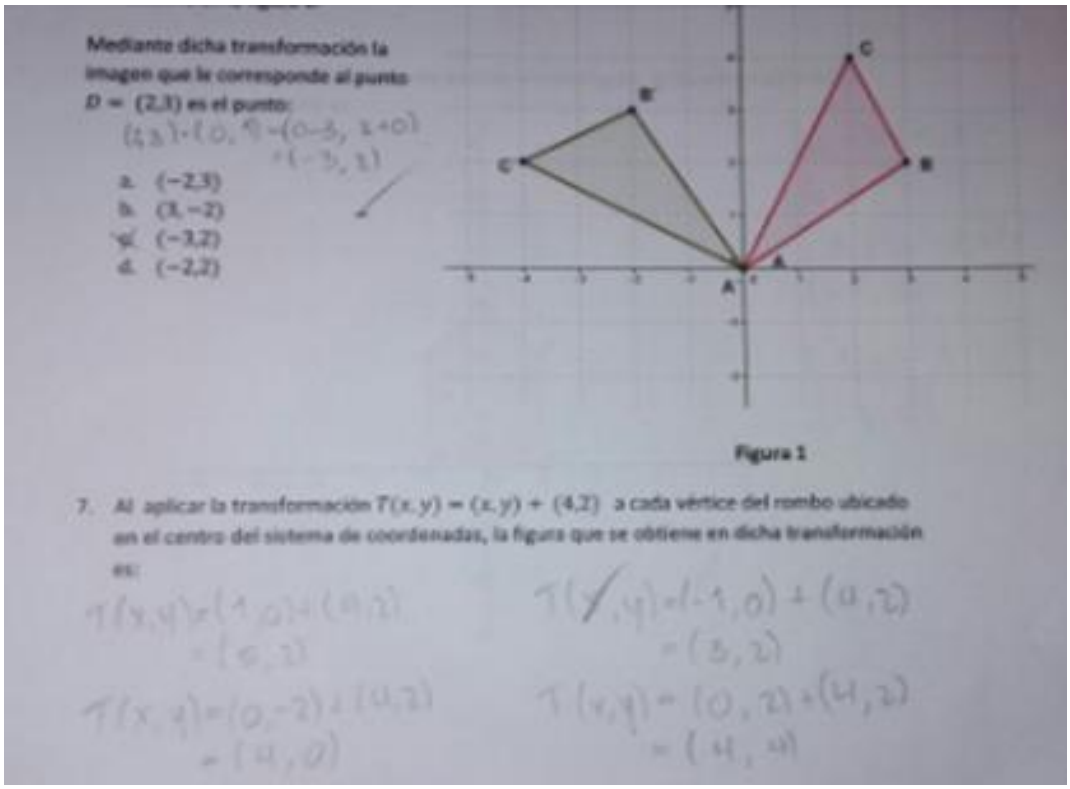


Figure 13. The photo of the construction of transformations was another activity carried out in the study

The students actively participated and enjoyed interactive games where the theory associated with complex numbers was applied. These games provided them with a practical and engaging way to explore and reinforce their understanding of complex numbers, including concepts such as addition, subtraction, multiplication, division, and graphical representation of the complex

plane. The interactive nature of the games made the learning experience enjoyable and helped solidify their knowledge of complex numbers.

The Enigma Game (Fig. 14) serves as the final step in the instructional process of this project, serving a dual role. First, it facilitates meaningful learning of the concept of complex numbers through the execution of videos, audio, and text readings that provide necessary information on the topic. Second, the game serves as an evaluative tool, offering various challenges through which students can self-assess their level of understanding of complex numbers. This approach makes evaluation a formative act, as students can revisit the informational resources to clarify any doubts they may have. When students feel fully competent, they can access a Google Drive tool where a closed-ended assessment instrument has been published.

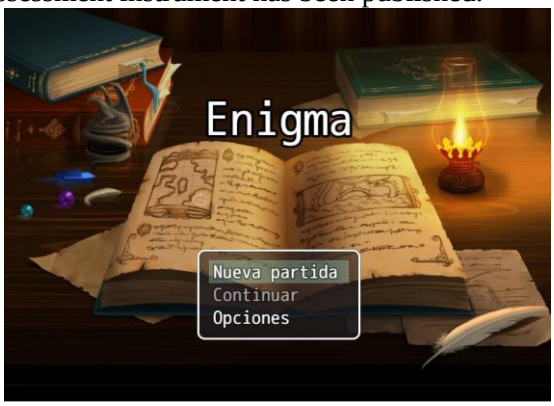



Figure 14. The screenshot of “Enigma Game”

### G. Some Remarks

Based on our research project and the observations made, it is important to mention the following difficulties encountered:

1. Challenges in understanding and applying transformations to transition from one plane to another.

2. Difficulty in solving new problems, as evidenced in the a-didactic situations.

3. The ease and enthusiasm with which students engage in algorithmic and repetitive processes, as seen in the table. However, these difficulties gradually diminished as individual and group activities were carried out in the formulation and validation situations.

## VII. Conclusions

The completion of this research project has yielded significant theoretical contributions and practical implications for mathematics education. Based on the data analysis from students in eleventh and tenth grade and the survey of mathematics teachers in Dosquebradas, our proposal effectively addresses the challenges students encounter in learning complex variables and highlights the need for more emphasis on this crucial topic in school curricula.

The initial diagnosis of tenth- and eleventh-grade students revealed several areas of improvement in their understanding of complex numbers, including their operations and applications in real-life problems. Our developed didactic proposal proved successful in enhancing students' mastery of complex numbers while simultaneously reinforcing essential geometric and trigonometric concepts.

The integration of Geoplano and Geogebra software significantly enhanced the comprehension and visualization of complex numbers and their functions, providing students with valuable insights into transformational concepts. Additionally, the strategic use of tablets and computers played a vital role in supporting the proposal implementation and fostering student learning.

The introduction of the educational game "Enigma" served as an innovative and engaging element, promoting discovery learning and stimulating students' exploration of complex variable functions across various topics.

The results demonstrate the effectiveness of our proposal in tackling the difficulties associated with learning complex numbers and underscore the potential of interactive and technology-driven approaches in mathematics education.

As a result, this research contributes new knowledge to the domain by presenting a comprehensive and effective didactic strategy to teach complex numbers. The findings not only inspire the academic community but also motivate practical engagement and scholarship among educators, encouraging them to adopt innovative teaching methods to empower students' understanding of complex variables and their applications in mathematics and beyond.

### A. Recommendations

After implementing the didactic proposal, the following recommendations should be considered for future research:

1. The study of complex numbers, from a geometric perspective, can be taught and understood by students at all levels of school through didactic instruments. This approach can lead them to comprehend multiple natural phenomena that are difficult to understand with real numbers. One example is the construction of fractals using complex variables.

2. The use of information and communication technologies (ICTs) is crucial in mathematics teaching and learning. ICT tools can serve as

motivating resources for student learning, offering scalability of concepts, the use of platforms that simulate reality, and formative assessment of content.

3. Further research is needed on complex numbers in schools, as they are a tool that facilitates the assimilation of not only the concept of function but also other mathematical topics such as limits, derivatives, integrals, sequences, series, and many other modern mathematical concepts.

4. Strengthening teacher research groups on topics related to complex variables is necessary to implement them in schools. This will improve students' performance levels and the competencies of future professionals.

By considering these recommendations, future research can continue to enhance the teaching and learning of complex numbers, promote the use of technology in mathematics education, and foster the development of mathematical competencies among students.

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